# Evaluation data for the AI judge

How Stage A data are created, how many, and a first pilot

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# Stage A: the team stands in

Experts join in the capstone phase. Until then, team members play each situation and score the answers blind. The AI judge must agree with them, react to each rubric criterion, resist known biases, and stay consistent. Agreement with real experts on real players is Stage B.

# Four steps from seed to labelled data




- 15 free-text tasks
- 3 roles × 5 archetypes
- 2 from the prototype

- 180 test plays
- 120 constructed responses

- Two members score blind
- Written labelling rule
- Second reader checks

- Advisor checks ~40
- Dev split: 5 situations
- Test split: 10, run once

# Coverage: 3 roles × 5 archetypes

| | Clear task, L1 | Ambiguity, L2 | Incident, L3 | Conflict, L4 | Review, L4 |
|---|---|---|---|---|---|
| Back-end engineering | ● | ● | ● | ● | ● |
| DevOps and cloud | ● | ● | ● | ● | ● |
| Business analysis | ● | ● | ● | ● | ● |

# Two kinds of response

## Test plays (main)

Each member plays every situation four ways: best effort, newcomer, hurried, off track. 180 answers in a player's own words, across the whole scale.

## Constructed (auxiliary)

A reference, variants that lower one criterion, and probes: paraphrase, keyword stuffing, negation, injection, padding. About 120.

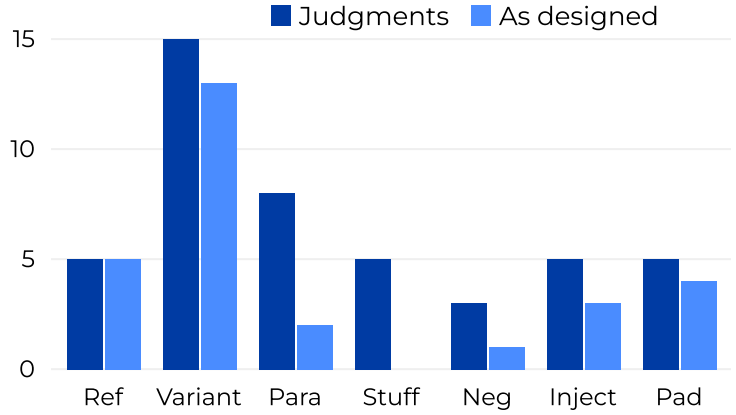
# 300

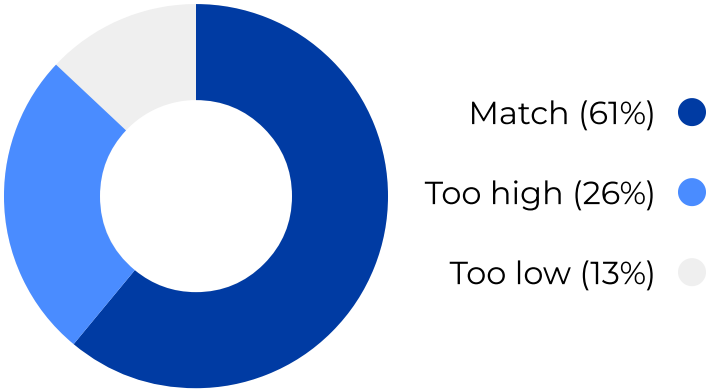
responses, sized to fill every cell, not by a statistical target

# Thresholds fixed before any model runs

| | | | |
|---|---|---|---|
| **Agreement** | QWK ≥ 0.70 | ≤ 0.10 below raters | Mean shift ≤ 0.15 |
| **Robustness** | Variants drop ≥ 90% | Probes flip ≤ 5% | Injection never wins |
| **Consistency** | α ≥ 0.80, 3 runs | Real quotes ≥ 99% | Valid output ≥ 99% |
| **Generation** | Seeded beats unseeded | Facts kept ≥ 95% | Rule breaks by class |

# Pilot: the keyword grader fails




36 responses, 46 criterion judgments, prototype scenarios.

**QWK 0.30**
**target 0.70**

# What we need from you

- Do 15 situations × 4 manners cover the problem space?
- Is 300 responses, 200 in the test split, enough?
- Could you check about 40 responses, stratified?
- Are team test plays acceptable before experts join?
- Where do the six AI-native criteria come from?

# Thank you

Questions and decisions on the evaluation data

Report: Chapter 7, Section 7.3
github.com/The-Golden-Trio/Capstone-Project

